SALMAN AKHATAR

Senior Data Scientist • AI/ML Engineer

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PROFESSIONAL SUMMARY

Senior Data Scientist / AI-ML Engineer with 9+ years of experience spanning backend engineering, full-stack development, and production GenAI systems. Proven track record architecting agentic AI workflows, LLM-powered enterprise tools, and cybersecurity automation platforms deployed at scale (20,000+ machines). Strong leadership history across client delivery, mentoring, and cross-functional program management, with recognition including a 2026 IBM Tech award for top-tier technical talent.

TECHNICAL SKILLS

Languages & Frameworks: Python, JavaScript, FastAPI, Django, Flask, ORMs (Django, SQLAlchemy)

AI / GenAI / ML: LLM, GenAI, RAG, Vector DB, Prompt Engineering, NLP, Machine Learning, LangChain, AutoGen

Data Engineering: PostgreSQL, Elasticsearch, PySpark, Pandas, Kafka, Confluent, Airflow, Celery

Infrastructure / DevOps: Docker, Podman, Kubernetes, Microservices Architecture, CI/CD, AWS / Azure / GCP

Other: RESTful APIs, Web Scraping, Git

WORK EXPERIENCE

Senior Data Scientist: AI/ML Engineer — IBM *Feb 2025 – Present*

- Selected as a Global IBM Tech 2026 awardee, recognizing top-tier technical talent; invited to collaborate with IBM's senior technical community in San Diego on enterprise AI architecture.
- Architecting end-to-end agentic AI workflows using LangChain and AutoGen, reducing manual processing load for core business units.
- Leading development of high-performance backends with FastAPI and Django, integrating LLM-based solutions with enterprise-grade security and governance controls.
- Driving WatsonX.ai-aligned integrations and multi-model orchestration across Granite, Llama, and related enterprise model stacks.
- Designing and optimizing scalable data pipelines with Confluent and PostgreSQL, reducing query latency for real-time AI workloads.
- Mentoring engineers and aligning technical strategy across high-stakes programs.

Senior Consultant — EY LLP *Nov 2021 – Feb 2025*

- Led end-to-end development and deployment of SAM and cybersecurity solutions, architecting cross-functional projects from requirements through delivery.
- Built and deployed two LLM-powered chatbot wrappers supporting client automation initiatives.
- Developed BOT software for automated, cross-platform data extraction, deployed across 20,000+ machines; cut manual data-handling time by 40%.
- Designed high-performance ETL pipelines processing millions of PostgreSQL records via FastAPI; boosted processing speed through Redis integration.
- Implemented RBAC and token-based authentication for secure API middleware access.
- Managed Docker-based deployments and supervised CI/CD pipelines on Azure VMs across dev, QA, and UAT environments.
- Promoted to Senior Consultant (Q4 2023); recognized with “Collaborative Extraordinaire” (Q2 2024) and “I Am Exceptional” (Q1 2023) awards.

Software Engineer III — TELUS International *Dec 2018 – Nov 2021*

- Engineered 5 high-impact Python applications using Flask and Django, automating data workflows to streamline operations.
- Built ETL pipelines with data preprocessing for BigQuery and automated workflows via Airflow, improving pipeline efficiency and data accessibility.
- Partnered with architects and analysts on end-to-end delivery; established best practices for quality, cost-effective execution.
- Contributed to product roadmap strategy in consultation with senior leadership. Awarded "Pat on the Back" for outstanding performance (Q2 2019).

Senior Software Engineer — App & Web Solutions Pvt Ltd *Jan 2018 – Dec 2018*

- Led development and integration across 5 projects, including a CNN-based image processing application.
- Built Python and PHP APIs supporting both web and mobile applications; developed UI schema, reports, and dashboards for data visualization.

Senior Software Engineer — Roundone Network Pvt Ltd *May 2017 – Jan 2018*

- Designed and led development for 2 key projects (including CRM panels) and several sub-modules, improving productivity and reporting efficiency by 85%.
- Promoted from Software Engineer to Senior Software Engineer within 2 months for exceptional performance.
- Built data crawlers for job-listing acquisition with multi-level preprocessing for platform use; led a team of 5–7 engineers while managing personal delivery responsibilities.

Software Engineer — Piana IT Solutions Pvt Ltd *Jan 2016 – May 2017*

- Contributed to full-stack development across 4+ projects and sub-modules, partnering directly with CEOs and stakeholders on technical specifications.
- Authored, tested, and debugged production code following full SDLC best practices.

NOTABLE PROJECTS

IBM — Super Agents

- LLM-enabled super agents that coordinate multiple single-purpose agents into a dynamic multi-agent flow to solve end-to-end problems.

IBM — Evaluation Agents

- Dynamic evaluation agent leveraging Llama and Claude (Sonnet) model families to score agent outputs against user-defined criteria such as accuracy, consistency, and coherence.
- Incorporated configured ground-truth references to validate agent responses, enabling automated, criteria-driven quality checks across multi-agent systems.

IBM — Staffing Builder for Consulting Team

- Studied existing estimation and staffing workflows within IBM Consulting and translated them into an agent-based tool for automated staffing recommendations.
- *Outcome: Significantly reduced manual estimation and staffing effort for the Solutioning team.*

EY Atom Bot & Atom Bot Server — Enterprise System Monitor

- Developed and deployed an automation bot across 25,000+ machines and a scalable backend server handling thousands of simultaneous requests, with robust ETL pipelines and full unit-test coverage.
- Technologies: Python3+, ETL Pipelines, API Design, PostgreSQL, Microservices, Docker/Podman, Azure, ReactJS.
- *Outcome: Reduced manual data-handling time by 40% across systems.*

EY Atom ETL Engine

- Automated ETL engine ingesting and transforming data from 15+ connectors into PostgreSQL, seamlessly integrated with EY Atom Bot.
- *Outcome: Processed millions of records daily at 99.9% accuracy; cut processing time by 70% with automated quality checks (95% issue detection).*

EY Quark — Cybersecurity Solution

- Scalable, microservices-based platform for agentless data discovery, application rollout, patch management, and configuration management, built on Flask/FastAPI with Ansible-driven compliance automation.
- *Outcome: Increased system uptime by 20% through automated misconfiguration remediation.*

EY Quasar — LLM-Powered Data Query & Visualization Tool

- LLM wrapper (Ollama, Mistral 7B) translating natural language into optimized SQL and auto-generated charts with summaries.
- *Outcome: Improved report generation time by 30% for non-technical users.*

EY Qbot — Document Intelligence System

- LLM-powered document Q&A system using ChromaDB for vector storage, returning context-aware answers with page-level source references.
- *Outcome: Indexed 1,000s of documents at 95% source-attribution accuracy; cut document query time by 70%.*

TELUS — Relevant Resume Finder

- NLP-based matching engine ranking 20,000+ resumes (PPT/DOCX/PDF) against job descriptions by relevance score.
- *Outcome: 90% relevance-matching accuracy; cut manual resume screening time by 85%.*

EDUCATION

- M.Sc. Tech, Industrial Maths with Computer Application — Jamia Millia Islamia, New Delhi (Jul 2013 – Jun 2016)
- B.Sc., Physical Science with Computer Science — University of Delhi (Jul 2010 – Jun 2013)

LANGUAGES

English, Arabic, Urdu, Hindi